

Section A – Network Verification and Baseline Understanding

1. List all devices in the home network and record

S/No	Device Name	Connection Type	IP Address
1.	Sever0(Server-PT)	Wired	192.168.0.117/24
2.	IOT 2(Light)	Wireless	192.168.0.101/24
3.	Laptop0 (Laptop-PT)	Wireless	192.168.0.106/24
4.	Printer0 (Printer-PT)	Wired	192.168.0.107/24
5.	PC0 (PC-PT)	Wired	No IP Assigned (Static IP Configuration)
6.	MyHomeRouter(Linksys-WRT300N)	Wired and Wireless	192.168.0.1/24
7.	AP&TISP(Cloud PT)	Wired	IP Not required
8.	TV0(TV PT)	Wired	IP Not Required
9.	IOT0 (Ceiling Fan)	Wireless	192.168.0.102/24
10.	IOT1 (Garage Door)	Wireless	192.168.0.103/24
11.	Laptop1(Laptop-PT)	Wireless	192.168.0.104/24

Screenshots:



Figure 1: Image of all devices in the home network showing IP address, Device name, connection type e.t.c

2. Identify the **default gateway** used by all home devices and explain why this device is a critical dependency.

The default gateway for systems with IP addresses “**192.168.0.1**” This is because all devices belong to the same 192.168.0.0/24 Class C network, and the default gateway must be the IP address of the router, which is responsible for forwarding traffic destined for the external networks AT&TISP, MyHomeRouter is single exit point from the Home network, therefore it IP address 192.168.0.1 is the default Gateway for all the devices.

3. From a laptop, verify reachability to the following and note the result: Desktop PC, IoT Server, One IoT device, ISP Cloud.

Reachability verification table for both Laptop0 and Laptop1

s/no	Devices	Reachability	Observation
1.	Desktop PC(0)	Not Reachable	No IP Address Assigned
2.	IOT Server	Reachable	Respond to Broadcast Ping
3.	One IOT Device	Reachable	All IOT devices Respond to broadcast Ping
4	ISP Cloud	Not Reachable	After sending a multicast the Router returned destination host unreachable

Explanation and Screen shoots

A single broadcast ICMP echo request was sent using the command, “ping 192.168.0.255 -n 1” This broadcast message was delivered to all devices within the home network. The results showed that multiple IoT devices responded, confirming that they were reachable and correctly connected to the network see figure 2.

However, PC0 (Desktop PC) did not respond to the broadcast because it had no IP address assigned, making it incapable of participating in IP-based communication. As a result, the Desktop PC was confirmed to be not reachable at the network layer. To further confirm the reachability status of devices, the ARP cache was examined using “arp -a” The ARP table displayed IP-to-MAC address mappings for reachable devices on the network, this is because the Mac address of PC0 connection was shown in the information card of PC0. The absence of an IP address associated with the MAC address of PC0 confirmed that it was not reachable, as ARP requires a valid IP address for resolution see figure 3

To test for the reachability of the ISP a unicast ICMP test was performed using a public IP address “ ping 8.8.8.8 “ this is because the ISP Cloud does not belong to the home network local subnet and cannot be reached using broadcast messages, as broadcasts are not forwarded by routers. The response received was “Reply from 192.168.0.1: Destination host unreachable.” laptop successfully communicated with the default gateway (MyHomeRouter), but the router was unable to forward the packet to the ISP Cloud. Therefore, the ISP Cloud was determined to be not reachable at the time of testing as shown in figure 4.

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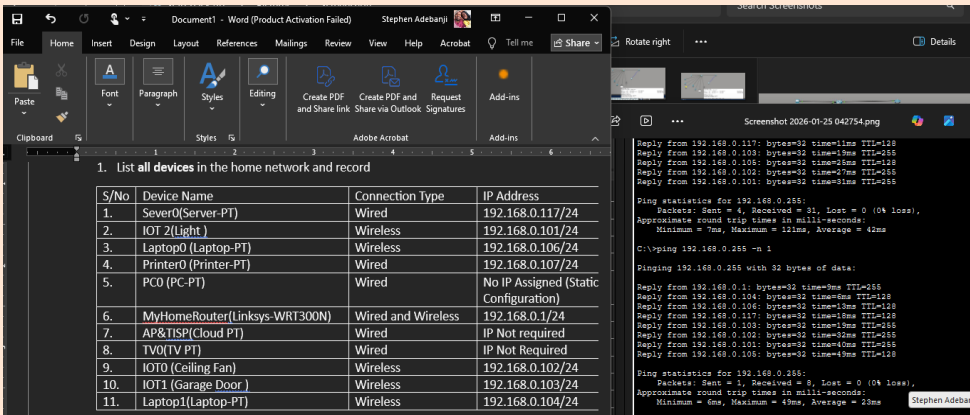


Figure 2: Image of the broadcast message sent from Laptop0, and comparison with the Table of question 1

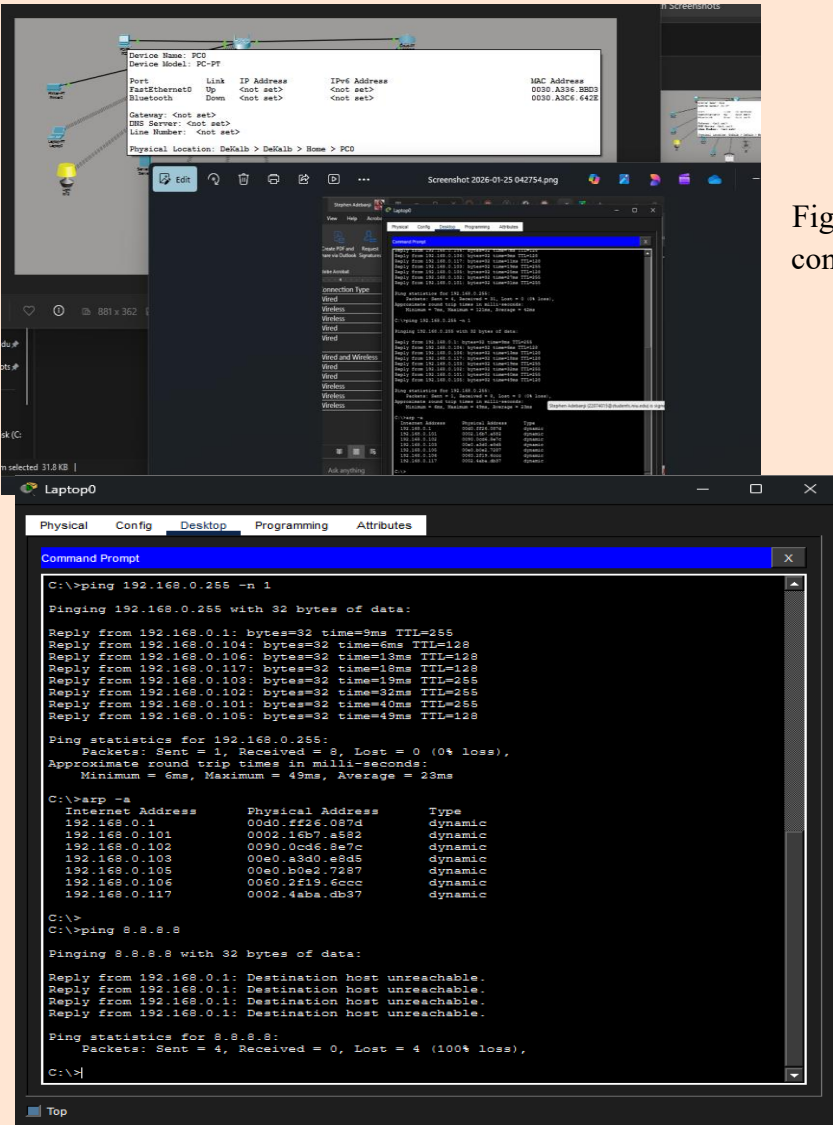


Figure 3: Image of ARP examination to confirm that PC0 was not reachable by laptop0

Figure 4: Image of multicast examine the reachability of the ISP cloud which was not reachable by laptop0

Section B – Attack Surface Identification

4. Based on the Logical view, identify the **single trust boundary** between the home network and the ISP. Explain what devices exist on each side of this boundary.

The single trust boundary is at MyHomeRouter, which separate the trusted home network devices from the untrusted ISP and internet infrastructure, this so because the MyHomeRouter is the single exit gateway for devices in the network to the remote ISP, this is the point where we can apply different security policies to each side, set up the firewall rules, access control and monitor the network.

At the trusted side of the boundary the devices that exist are Sever0(Server-PT), IOT 2(Light), Laptop0 (Laptop-PT), Printer0 (Printer-PT), PC0 (PC-PT), MyHomeRouter(Linksys-WRT300N), IOT0 (Ceiling Fan), IOT1 (Garage Door), Laptop1(Laptop-PT) and at the untrusted side which is we have AP&TISP(Cloud PT) and TV0(TV PT).

5. Using the Physical view, identify **three devices** that are: Always powered on, and Rarely interacted with directly by users. Explain why these characteristics increase their attractiveness to attackers.

The three devices that are always powered on are:

- (a) MyHomeRouter (Linksys-WRT300N): MyHomeRouter must remain continuously powered on to provide internet access to all devices on the network. It is rarely interacted with directly by users, as it is typically configured once and then left unattended, with infrequent access to its administrative interface. These increased its attractiveness to attackers as it is at the trust boundary between the home and the ISP, if MyHomeRouter is compromised by attacker, the attacker can monitor traffic, redirect DNS and devices in both trusted and untrusted side of the network.
- (b) AP&TISP (Cloud PT): The AP&TISP (Cloud PT) is always powered on in order to preserve client connectivity, the ISP infrastructure devices run continuously, it is rarely interacted with directly by the users as users can not interact with this device directly, only ISP technicians are responsible for managing it. This is attractive to the because it also handles traffic for many customers not just one home, if it is compromised by attacker, it will affect data confidentiality, traffic can be routed to another server by attacker and service to the ISP can be denied, this device is a very high-level value target for large scale attacks.
- (c) Sever0(Server-PT): Which is in AT&T ISP Building, must always be powered on to continuously deliver services to the IOT devices in the home network, it is rarely interacted with directly by the user as it is managed remotely by administrators at the AT & TISP building with no daily physical user interaction. Due to the fact that this device is always on it allows the attackers to have more time to probe its vulnerabilities, successful compromise of this will enable the attacker to be persistent and have smooth lateral movement to attack the whole network, this is attractive to attackers as servers usually store sensitive data, authentication services, logs and configuration of the system.

6. Identify which devices in the home network: Store user data, Control physical systems, Provide centralized services
- (a) Devices in the home network that stores user data are Laptop0, laptop1, PC0, they store personal files, documents, credentials, browsing history, applications data, they also contain most sensitive user data such as logins, emails and saved password if compromised attackers will gain direct access to user information and identities.
 - (b) Devices in the home network that controls physical systems are IOT2(Light) this device interact with the physical system by turning the light on/off, IOT0(Ceiling Fan) interact with the physical system by controlling the airflow and IOT1(Garage Door) control the physical system by closing and opening the garage door, compromise of this system can lead to physical intrusion risk, when garage doors are open remotely, safety and comfort issues by disabling the light, attacker can also carry out surveillance and this devices can also become botnet, in all compromise of this devices will become a very serious physical security risk.
 - (c) Devices in home network that provide centralized services are , MyHomeRouter which provides centralized network services including routing, IP address assignment and internet access, Server0 this provide centralized IOT management services coordinating communication, control logic and data exchange between the IOT devices, and lastly the printer which offers shared printing services to multiple users on the network, compromise this devices can lead to full visibility of network traffic, user redirection to malicious websites (ie DNS Poisoning) Bypass of internal security controls, it can be catastrophic if the attacker compromise the network through this device, also through the server0 attacker can again coordinated control into the physical system, use any of the IoT devices as a persistent foothold inside the network, manipulate IoT data, inject malicious command into the system and even gain centralized control of the system. The printer is often overlooked but is exploitable the attacker can use it as an internal pivot point for lateral movement, carry out interception of printed documents if the printer is compromised.

Section C – IoT Server and Centralized Trust

7. Log in to the IoT Server using: **Server-PT → Desktop → IoT Monitor**, List all IoT devices shown in the IoT Monitor interface.

The Devices in the monitor are IOT0(Ceiling Fan), IOT1(Garage Door) and IOT2(Light) see figure 5

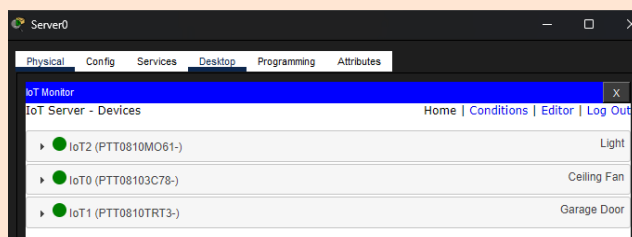


Figure 5: Image of Serve0 IOT monitor screen on Desktop

- Toggle the state of **at least two IoT devices** (e.g., lamp, fan, garage door). Based on this observation, explain what this reveals about: Centralized control, Trust relationships between the server and IoT devices.

Base on the observation the prompt response of IoT devices despite a remote server reveals centralized control and strong trust relationships between the server and IOT devices. While this enables efficient automation, it also introduces significant security risks, as compromise of the centralized server or its communication channel can lead to total loss of control over physical systems in the home network. See figure 6

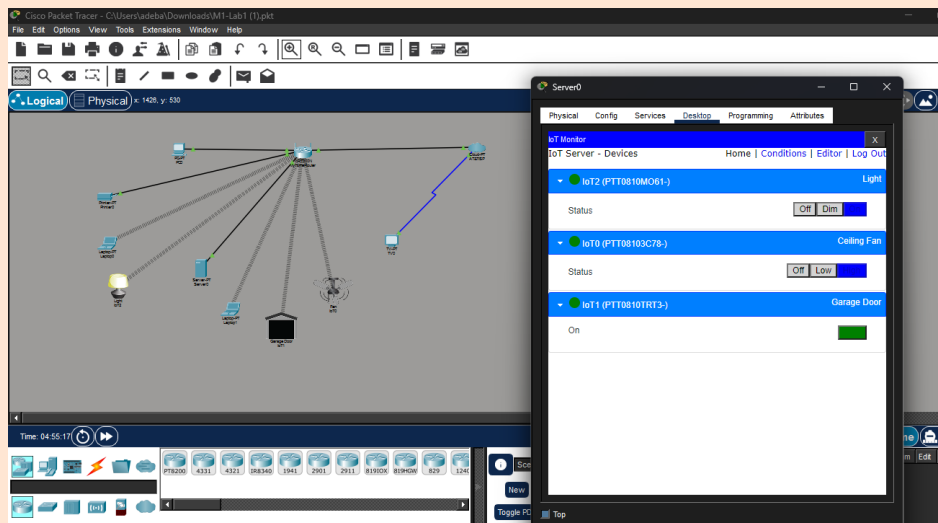


Figure 6: Image of Toggle the state of at least two IoT devices

- Explain why the IoT Server represents a **high-value internal target**, even though it is not directly exposed to the Internet.

The IOT Server represent a High value internal target, even though is not directly exposed to the internet because, the server provides centralized management and control for multiple IoT devices, which implicitly trust it to issue valid commands. The server operates continuously with minimal direct user interaction and aggregates important control logic and telemetry data, making it an attractive target for attackers.

If the IoT server were compromised, an attacker would be able to centrally manipulate physical devices, such as lights, fans, and the garage door, without interacting with each device individually. In addition, the compromised server could be used for lateral movement within the home network and to maintain persistent, low-visibility access. Overall, this would result in a loss of control, integrity, and safety of the IoT system, significantly weakening the security of the entire home network.

Section D – Lateral Movement Scenarios

10. Assume **Laptop0** is compromised by malware. Using observed reachability, list **all devices** the attacker could access **without exploiting additional vulnerabilities**.

When Laptop0 is compromised, it becomes the attacker's initial foothold inside the trusted home network. Because Laptop0 is a legitimate internal device with a valid IP address, it is allowed to communicate freely with other systems on the trusted side of the network without triggering security controls. Using a broadcast ICMP echo request, the attacker will be able to perform network discovery, identify active hosts on the network. This confirms which devices are reachable and which are not, effectively revealing the internal network layout.

Based on the observed reachability, the attacker can directly access the IoT server, the IoT devices, and the home router at the trust boundary. The IoT server is a particularly attractive lateral movement target because it is always powered on, it centrally manages communication with multiple IoT devices, and is implicitly trusted by them. By pivoting from Laptop0 to the IoT server, the attacker moves from a single compromised endpoint to a centralized internal system, significantly increasing their level of control and persistence within the network.

From either the compromised Laptop0 or the IoT server, the attacker can then move laterally to the IoT devices, such as the light, ceiling fan, and garage door. These devices accept commands from trusted internal sources and do not independently validate command intent, allowing the attacker to manipulate physical systems directly. This represents a critical escalation, as the attack transitions from purely digital compromise to physical impact, including unauthorized control of household systems.

In addition, the attacker can also reach MyHomeRouter, which represents the trust boundary between the home network and the ISP. Communication with the router is confirmed by ICMP responses returned during reachability testing. While the attacker cannot reach the ISP Cloud, access to the router allows for reconnaissance of gateway behavior and positions the attacker close to the network's control point, increasing visibility into internal traffic and potentially enabling further attacks if weaknesses exist.

Importantly, lateral movement in the network is limited to devices within the trusted network and the trust boundary. The desktop PC (PC0) remains unreachable due to the absence of an IP address, and the ISP Cloud is not reachable because routers do not forward broadcast traffic and were unable to route unicast traffic externally at the time of testing. Overall, this lateral movement path demonstrates that basic internal reachability and implicit trust are sufficient to enable significant attacker movement, even without exploiting additional vulnerabilities, highlighting the security risks of flat network designs in home IoT environments.

11. Assume the **table lamp IoT device** is compromised. Identify which systems become reachable and explain why this is possible.

When the table lamp IoT2 (Light) device is compromised, it becomes an attacker's internal entry point into the trusted home network. Because IoT2 has a valid IP address and operates on the same trusted subnet as other devices, it can communicate freely within the network without needing to bypass additional security controls. This implicit trust allows the attacker to initiate lateral communication from the compromised IoT device.

From IoT2, the attacker can directly reach the IoT server (Server0) because the server is designed to centrally manage and communicate with all IoT devices. This trusted, always-on communication channel makes the IoT server a natural pivot point. Once reached, the server can be used to coordinate further actions and significantly amplify the attacker's control over the IoT environment.

The attacker can also reach other IoT devices (such as the fan and garage door) and the home router (MyHomeRouter), since all reside within the same trusted network and rely on the router as the default gateway. However, devices without IP addresses, like the desktop PC, and external systems such as the ISP Cloud remain unreachable. This demonstrates that compromising a single IoT device can expose centralized services and physical systems due to shared network access and implicit trust within the home network.

12. Explain how the flat network design enables lateral movement in both scenarios above.

The flat network design used in the home network places user devices, IoT devices, the IoT server, and the router on the same trusted subnet with minimal internal segmentation. Because there are no internal trust boundaries or access controls between device categories, any device that successfully joins the network is treated as trusted. This allows compromised endpoints such as Laptop0 or the IoT2 table lamp to communicate freely with other internal systems without needing to bypass additional security mechanisms.

In the first scenario, where Laptop0 is compromised, the flat network allows the attacker to use simple broadcast and unicast communication to discover and reach internal assets such as the IoT server, IoT devices, and the home router. Since internal traffic is not restricted, the attacker can move laterally from a single user device to centralized services and physical systems purely based on network reachability and implicit trust.

Similarly, in the second scenario, when the IoT2 device is compromised, the same flat network design enables lateral movement from a low-capability IoT endpoint to higher-value systems. The IoT device can directly communicate with the IoT server, other IoT devices, and the router because all devices share the same subnet and security policy. In both cases, the absence of segmentation or internal filtering allows lateral movement without exploiting additional vulnerabilities, highlighting how flat network designs significantly increase the impact of a single-device compromise.

Section E – Availability and Impact Analysis

12. Identify **two single points of failure** in the home network and explain the impact if each fails.

The home router (MyHomeRouter) is an explicit single point of failure in the home network. IP addressing, internal routing, and access to external networks are all provided by the router, which serves as the main hub for all network communication. Every device in the home network loses communication to the Internet and to each other if the router malfunctions or is compromised, essentially resulting in a total network outage. Furthermore, a router failure eliminates the trust boundary's primary security control, raising the possibility of traffic interruption, visibility loss, or attack exposure.

The IoT server (Server0), which offers centralized management and control for every IoT device, represents a second single point of failure. IoT devices depend on the server for coordination and command processing, thus if the IoT server fails or is compromised, IoT devices won't be able to respond to control orders or operate properly. If the server is compromised, this could result in the loss of automation, the incapacity to manage physical systems, or illegal activity. When taken as a whole, these single points of failure show how, despite its convenience, centralization may have a substantial impact on a home network's availability, control, and security.

13. Describe what happens to: IoT device control, User device communication if the IoT Server becomes unavailable.

For IoT device control, devices such as the smart light, fan, and garage door lose their central command source. Since these devices rely on the IoT Server for receiving control instructions and automation logic, they may stop responding to remote commands or scheduled actions. In most cases, the devices will either remain in their last known state or operate only with very limited local functionality, resulting in a loss of centralized control and automation over physical systems.

For user device communication, devices such as laptops and PCs remain largely unaffected at the network level. User devices can still communicate with each other and access internal network services through the home router because their communication does not depend on the IoT Server. However, users lose the ability to monitor or control IoT devices through applications or dashboards that depend on the IoT Server, reducing usability but not basic network connectivity.

14. Which single device failure or compromise would result in the **greatest overall impact** on the home network? Justify your answer using availability, safety, and functionality considerations.

The single device whose failure or compromise would have the greatest overall impact on the home network is the home router (MyHomeRouter).

From an availability perspective, the router is the central gateway for all network communication. If it fails or is compromised, internal devices may no longer communicate with each other, and all connectivity to external networks is lost. This results in a complete network outage, affecting user devices, the IoT server, and all IoT devices simultaneously.

From a safety perspective, the router sits at the trust boundary between the home network and the ISP. A compromised router allows an attacker to intercept, modify, or redirect traffic, potentially enabling attacks against internal systems without detection. This can indirectly lead to unsafe conditions, such as unauthorized control of IoT devices or exposure of sensitive data, even if those devices themselves are not directly compromised.

From a functionality perspective, the router enables essential services such as IP address assignment (DHCP), routing, and traffic filtering. If the router is unavailable, IoT automation, user applications, and centralized services all fail to function properly. While failure of the IoT server impacts automation and physical device control, basic network communication can still continue. In contrast, failure or compromise of the router disrupts all network operations, making it the most critical single point of failure in the home network.

Section F – Physical vs Logical Perspective

15. In the Logical view, how many distinct networks or domains are visible?

In the Logical view, there are two distinct networks (or domains) visible. The first is the home network, which includes all trusted internal devices such as the laptops, desktop PC, IoT devices, the IoT server, printer, and the home router’s LAN side. These devices share the same local subnet and operate under a common trust domain, allowing direct internal communication.

The second is the ISP / Internet network, represented by the ISP Cloud and external networks beyond the home router. This domain is outside the homeowner’s control and is separated from the home network by the router, which acts as the trust boundary between the trusted internal network and the untrusted external network.

16. In the Physical view, how many devices actually exist within the home network domain?

In the Physical view, there are Ten (10) devices that actually exist within the home network domain. These devices are home router, which provides network connectivity, three user devices (PC0, Laptop0, and Laptop1), a printer, the power distribution device inside the rack though not connected to any device, three IoT devices (the smart light, fan, and garage door) and TV. All of these devices are physically located within the home network domain see figure 7

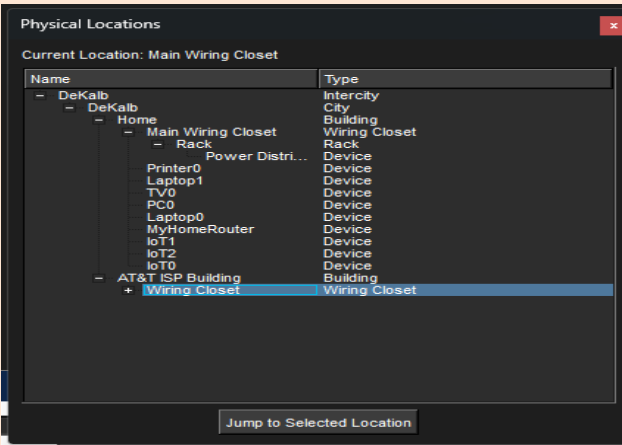


Figure 7: Image of devices at the physical view of the home network domain

18. Explain why attackers rely more on **logical reachability** than physical location when planning attacks.

Attackers rely more on logical reachability than physical location because cyber-attacks are conducted over networks, not through physical proximity. What matters to an attacker is whether a system can be reached, communicated with, and trusted at the network and application layers, regardless of where it is physically located. A device that is logically reachable can be targeted, scanned, and interacted with even if it is in another room, building, or country.

Logical reachability also determines the attack path. If a device shares the same subnet, trust domain, or routing path as a compromised system, an attacker can move laterally to it using normal network traffic. In contrast, physical location offers little protection if network access exists. For example, a remotely

located IoT server can still be controlled instantly if it is logically reachable and trusted, as observed in the lab.

Finally, attackers focus on logical reachability because it enables stealth, scale, and persistence. Attacks can be automated, repeated, and hidden within legitimate network traffic without any physical presence. As a result, systems that are logically reachable and implicitly trusted pose a much greater risk than those that are physically nearby but isolated at the network level.

Section G – Reflection and Preparation for Module 2

19. Identify **one architectural weakness** in this network that is not related to device software vulnerabilities.

One clear architectural weakness in this network, unrelated to device software vulnerabilities, is the flat network design with a single trust domain.

All user devices, IoT devices, the IoT server, and the router's internal interface reside on the same subnet and share the same level of trust, with no internal segmentation or access control between them. This means that once any single device is compromised whether a laptop or a low-security IoT device an attacker can move laterally to other systems based purely on network reachability, without needing to exploit additional vulnerabilities.

This architectural design amplifies the impact of compromise by allowing attackers to reach centralized services and physical control systems easily. The absence of internal trust boundaries, VLAN separation, or network-level filtering makes lateral movement trivial and significantly weakens the overall security posture of the home network.

20. Briefly explain how **network segmentation** could reduce risk in this topology, without proposing specific configurations.

Network segmentation would reduce risk in this topology by dividing the flat home network into multiple trust zones with controlled communication paths between them. Instead of all devices sharing the same level of trust, user devices, IoT devices, and centralized services would be logically separated, limiting which systems can directly communicate.

By enforcing boundaries between these segments, a compromise in one part of the network would be contained and less likely to spread laterally. For example, if an IoT device or a user laptop were compromised, segmentation would prevent it from freely reaching critical systems such as the IoT server or other devices, thereby reducing the overall impact and improving the security resilience of the network.